

Timewizer

A short guide to making your school timetable.

Timewizer works on your own computer and needs no internet. Each school is one file that you can save, copy and open again, like a document.

1. Install and open

Windows

Double-click the **setup .exe** file and follow the steps. Then open **Timewizer** from the Start menu. If Windows shows **"Windows protected your PC"**, press **More info**, then **Run anyway** (the installer is not signed yet). School IT staff can use the **.msi** file instead. Double-clicking a saved school file (`.twz`) opens it in Timewizer.

Ubuntu

On Ubuntu, double-click the **.deb** file and press Install (or run `sudo apt install ./Timewizer_0.2.0_amd64.deb` in a terminal). Then open **Timewizer** from the applications menu.

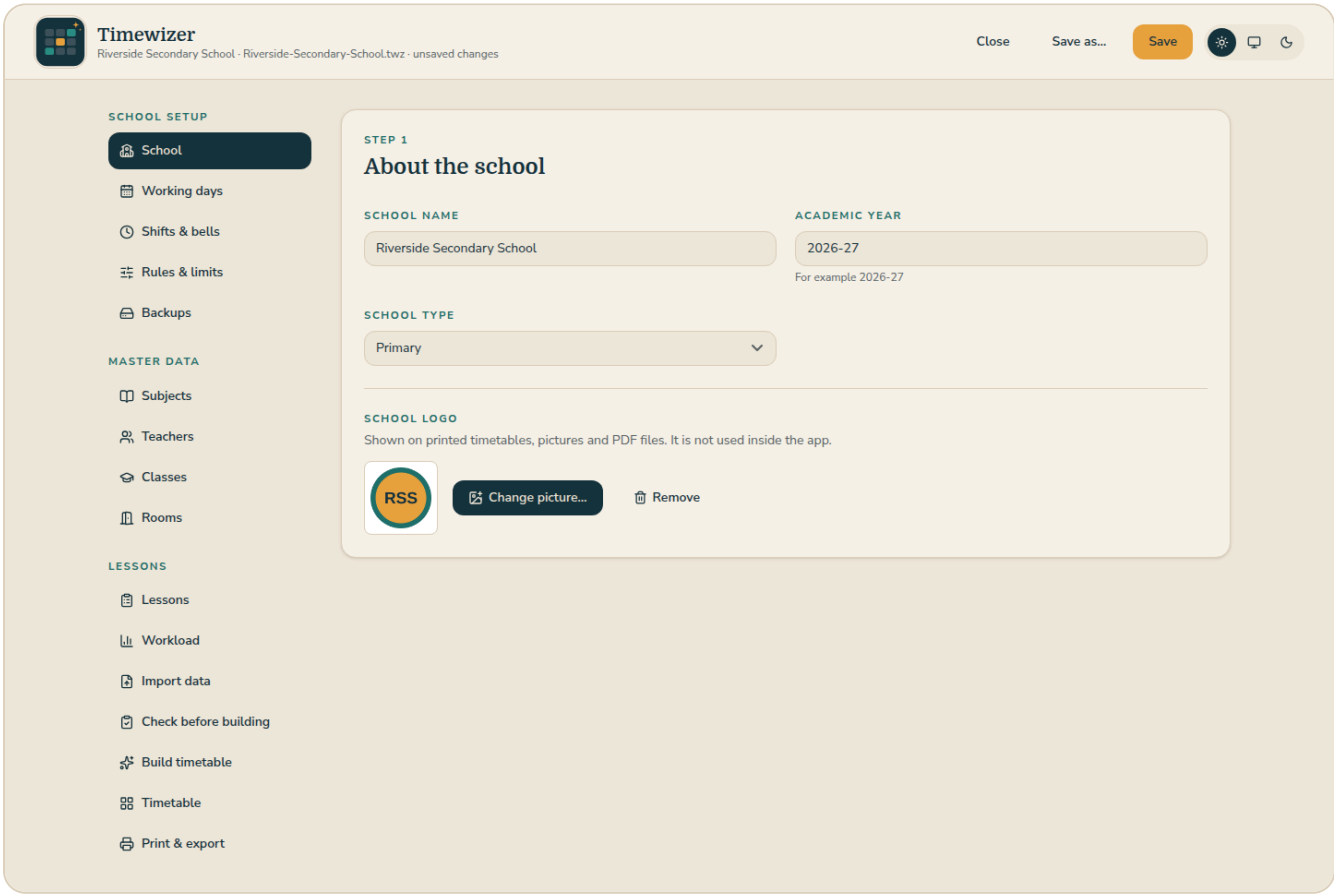
You can also use the **AppImage** file instead: right-click it, choose Properties, allow it to run as a program, then double-click it. Nothing is installed.

When Timewizer starts, a short animation plays (under two seconds). Click or press any key to skip it. To switch it off, open a school and go to **School setup > Preferences**. It is never shown when you open a school file by double-clicking it, or when the computer is set to reduce motion.

Start a school

1. Type the school's name and press **Create school**. Choose where to keep the school file.
2. Or press **Open file...** to continue with a school you saved before.

The steps are in the list on the left, from top to bottom. Press **Save** (or `Ctrl` + `S`) often. **Save as...** makes a copy, which is a good way to keep a backup.



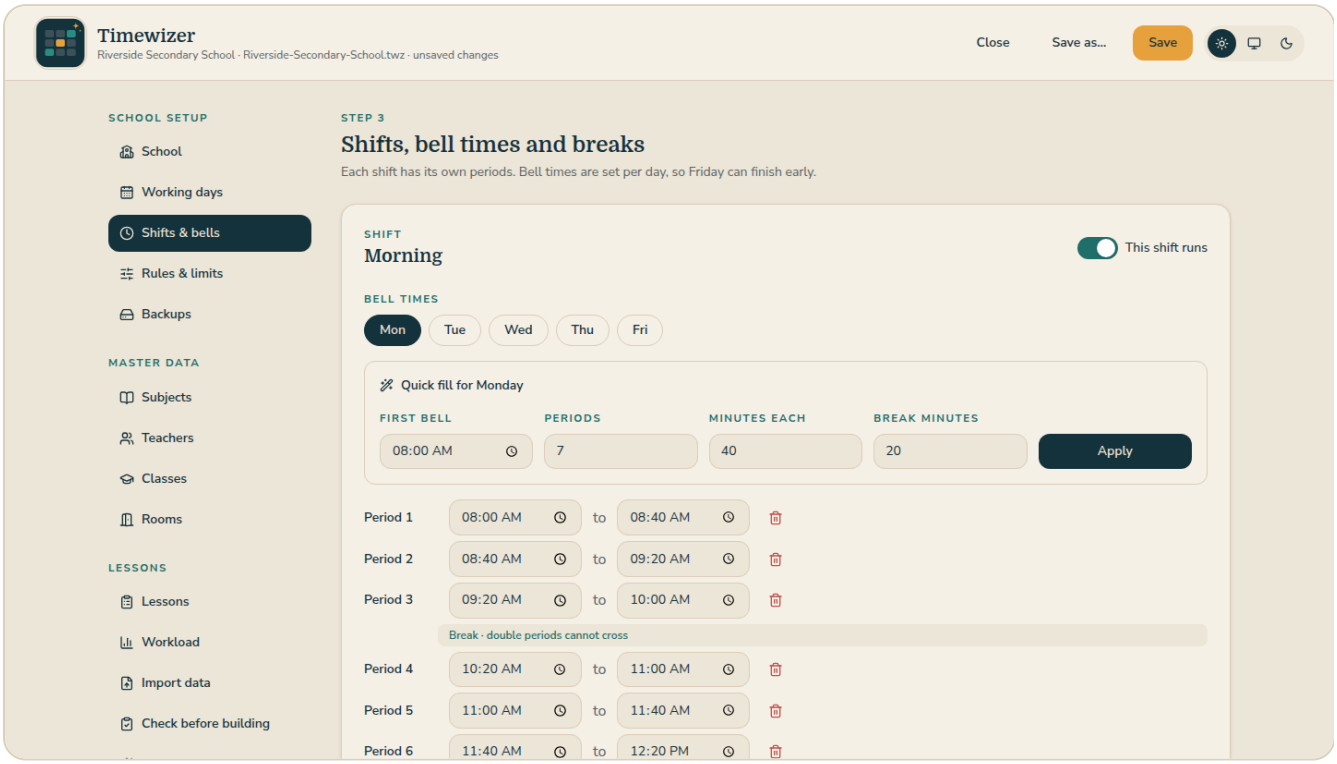
The School page. Add your school's own logo here; it is printed on timetables.

Want to look around first? The folder *samples* has a made-up school. Create a new school, open **Import data**, choose the five files in that folder and press Import.

2. Days, shifts and bell times

Working days: switch on the days your school runs lessons (Saturday is optional).

Shifts & bells: a school can have a morning shift, an afternoon shift, or both. Each shift has its own periods. Bell times are set for each day, so Friday can finish early. Use *Quick fill* to make a day's periods in one go, then *Copy Mon times* to the other days.



Break after period 3, switched on as "cannot be crossed" so double periods never run across it.

Rules & limits

School-wide rules, kept every time a timetable is built:

- **Teacher limits**: the most periods a teacher may teach per day and per week. A teacher can have their own numbers on the Teachers page.
- **Same subject on the same day**: normally 1, so a class never has two separate Maths sessions on one day.
- **What makes a timetable good**: sliders for free periods, spreading lessons across the week and so on. 0 means ignore, 10 means very important.
- **Bending rules**: if a full timetable is impossible, Timewizer can bend a few rules and tell you exactly which.

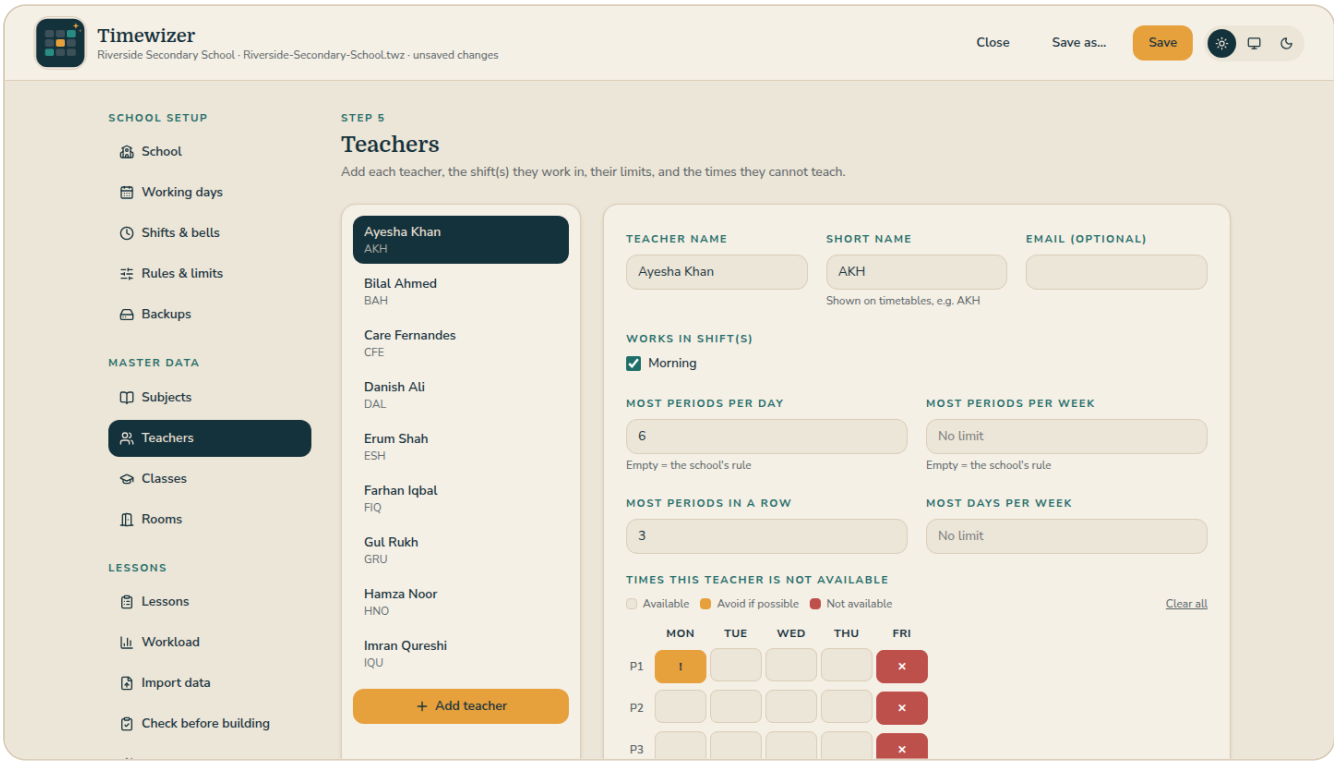
3. Subjects, teachers, classes and rooms

Add each of these under **Master data**. Pick one in the list, then fill in its details.

- **Subjects**: a name, a short name and a colour. The colour is used on timetables.
- **Teachers**: the shift or shifts they work, their limits, and the times they cannot teach.
- **Classes**: each belongs to one shift. A class can be split into groups (for example Boys and Girls).
- **Rooms**: a name and a size; rooms can have times they are closed.

Times people cannot teach

Click a box to cycle: ☐ available, 🟧 avoid if possible, 🟥 not available. Click a day's name to block the whole day. Striped boxes are periods that do not exist that day.



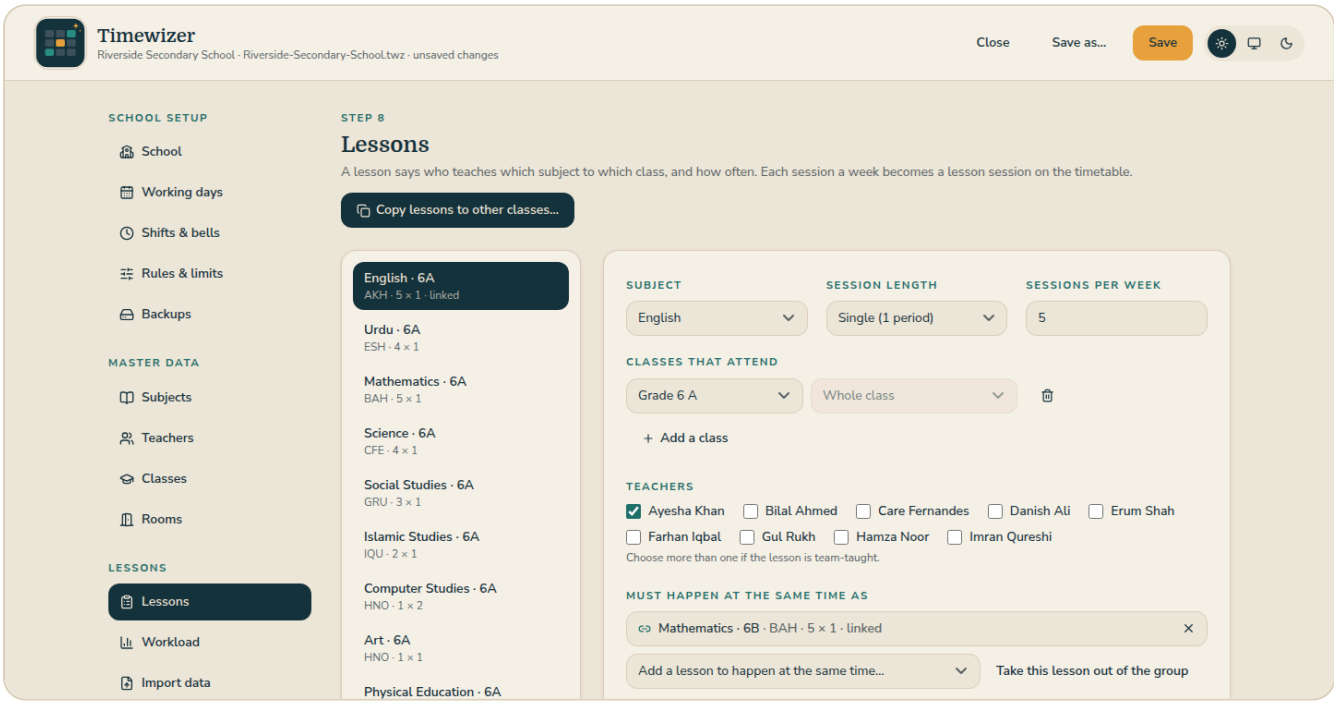
A teacher with Friday blocked and Monday period 1 marked "avoid if possible".

> **Faster:** use **Import data** to bring in subjects, teachers, rooms, classes and lessons from Excel or CSV. The page has downloadable templates. Rows that already exist are skipped, so importing twice is safe, and any row that cannot be used is listed with the reason.

4. Lessons

A **lesson** says who teaches which subject to which class, how many sessions a week, and how long each is (single, double or triple period). The words used everywhere in Timewizer:

Word	Meaning
Lesson	One row in the Lessons list, for example "English · 6A".
Lesson session	One occurrence of a lesson in the week. A double period is one session.
Period	One time slot. A double period is two periods.



A lesson with its teacher, rooms and a "same time" link.

Same time

Some lessons must happen together, for example two halves of a class swapping ICT and Library, or electives across sections. In the lesson, use **Must happen at the same time as**. Session 1 of each linked lesson is placed together, with its own room.

Copy lessons to other classes

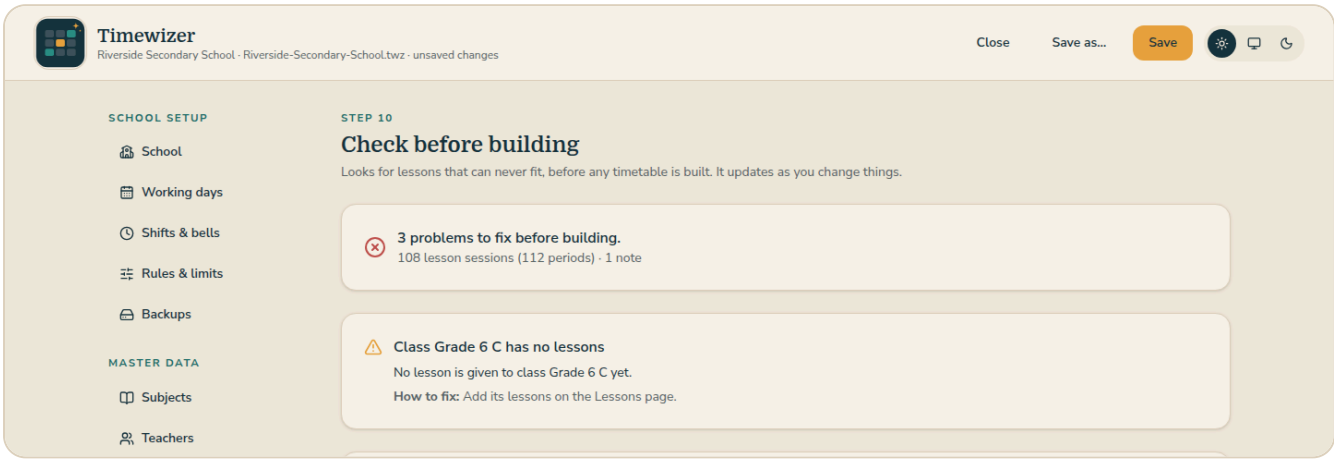
Press **Copy lessons to other classes...**, choose the class to copy from and the classes to copy to. A preview shows what will be added, and nothing changes until you press the button at the bottom. Then change the teachers if they differ.

Workload shows each class and teacher: periods used compared with periods available.

5. Check, then build

Check before building

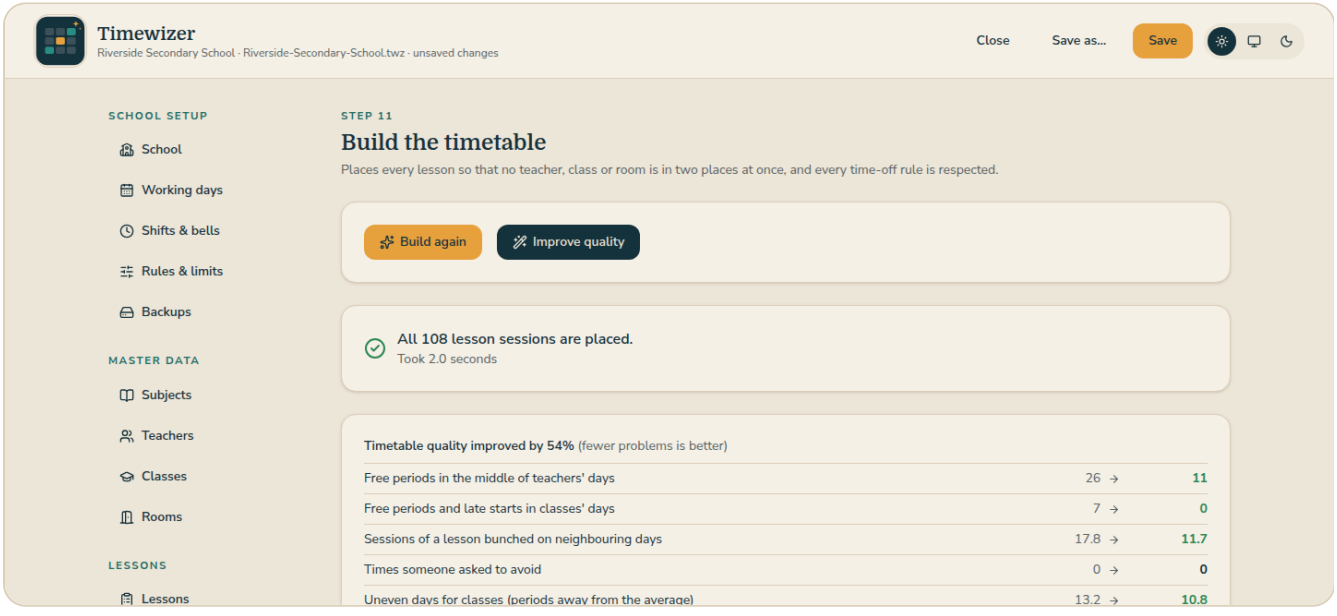
This finds lessons that can never fit, before anything is built: a teacher with too many periods, a lesson whose teacher and class are never free together, a room that is not big enough for the demand. Each problem has a plain explanation and **How to fix**.



Here three teachers have more periods than the weekly limit that was set.

Build timetable

Press **Build timetable**. A bar shows progress and you can press **Stop**. First all lesson sessions are placed, then the timetable is improved using the wishes from Rules & limits. The result shows how many sessions were placed, and how much better the timetable became.

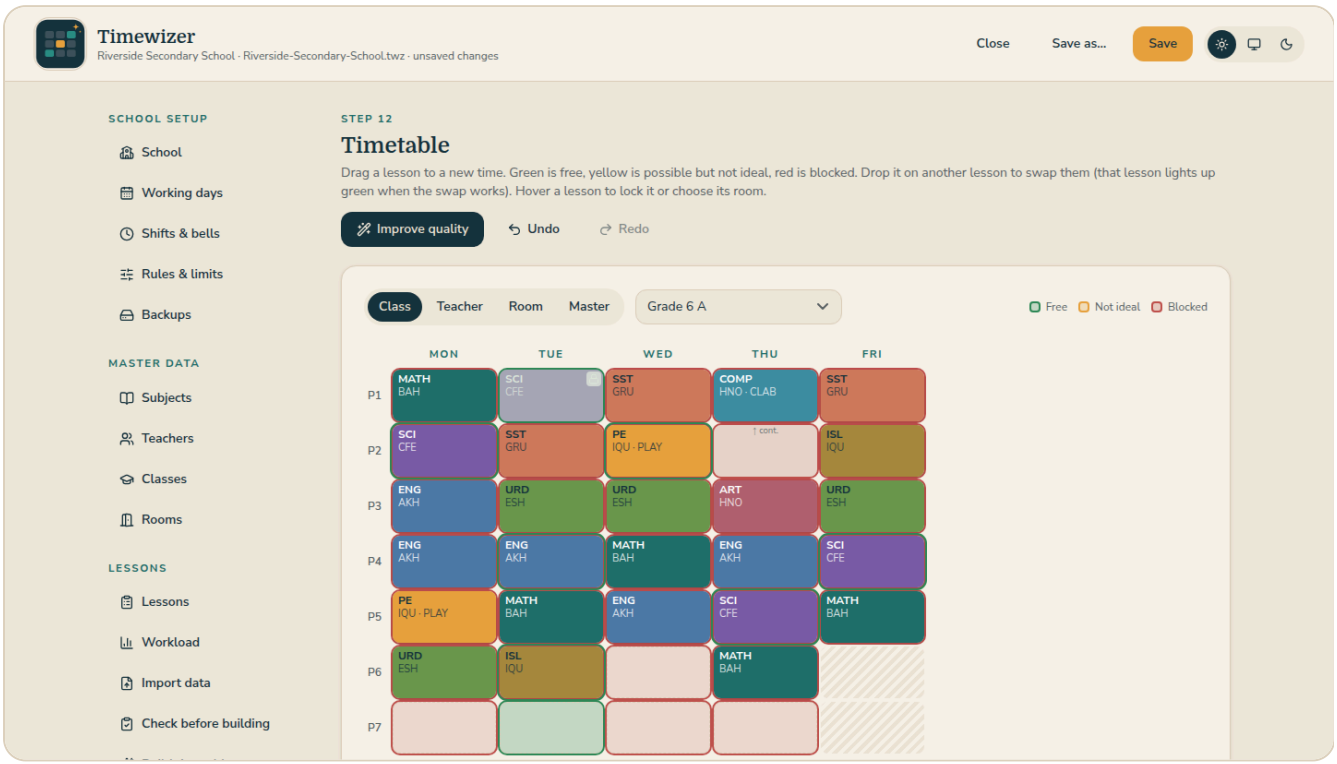


All lesson sessions placed, with the quality report.

If some sessions cannot be placed, each one is listed with the reason and a suggestion. If you switched on **bending a few rules**, the page lists every rule that was bent.

6. Looking at and changing the timetable

Open **Timetable**. Choose **Class**, **Teacher**, **Room** or **Master** (all classes together; use the zoom to make it bigger).



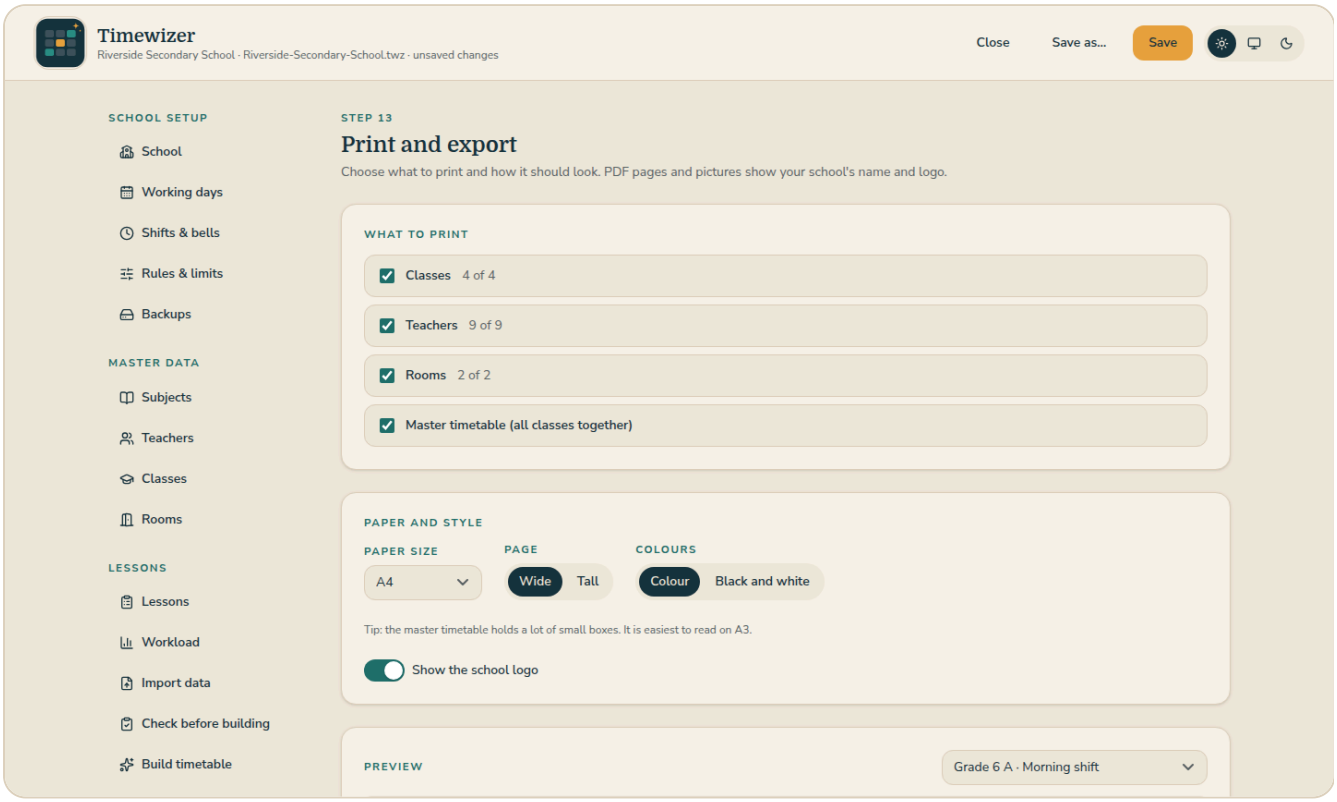
Dragging a lesson. Green boxes are free, amber is allowed but not ideal, red is a clash.

- **Move**: drag a lesson to a green or amber box. Hover any box to read why: red means a clash (the teacher, class or room is busy, or the teacher is not available); amber means it is allowed but breaks a "nice to have" rule from Rules & limits (an "avoid" time, a free period in the middle of someone's day, or two sessions of a lesson on days next to each other).
- **Swap**: drop a lesson on another one. If the swap works, that lesson lights up green.
- **Lock**: hover a lesson and press the padlock. A locked lesson shows a small padlock and cannot be dragged until you unlock it. Locked lessons stay exactly where they are when you build again, and the locks are saved in the school file.
- **Room**: hover a lesson and press the door to choose its room.
- **Take off**: drag a lesson to the box at the bottom; drag it back to place it again.
- **Undo / Redo**: the buttons at the top, or `Ctrl` + `Z` and `Ctrl` + `Y` (on the Timetable and Build timetable pages). Hover the button to see what it will undo. Building the timetable again counts as one change, so one Undo brings the old timetable back.
- **Change history**: below the timetable, a list of your changes, newest first, with the time and what was done (for example "Moved Science (6A) from Monday period 2 to Tuesday period 4"). Undone changes are crossed out until you redo them or make a new change. The list starts empty each time you open the school.
- **Improve quality**: tidies the timetable after your changes. Locked lessons never move.

> **If you change lessons or time off** after building, the page warns that the timetable may be out of date. Build again; your locked lessons stay put.

7. Print and export

Open **Print & export**. Choose what to print (classes, teachers, rooms, the master timetable, or just some names), the paper (A4, A3 or Letter, wide or tall), colour or black and white, and whether to show the school logo. A preview shows the page.



The Print & export page.

Button	What you get
Save PDF	One file, one page for each timetable, with your school's name and logo. Best for printing.
Save pictures (PNG)	One picture for each timetable, in a folder you choose. Good for WhatsApp or e-mail.
Save Excel	A workbook with the timetables you ticked: the master timetable, and sheets for classes, teachers and rooms, plus a list of every lesson session. Each period shows its times; if a day has different bell times (for example Friday), they are shown too. Excel does not need to be installed.
Save CSV	The timetables you ticked in one CSV file (opens in Excel or LibreOffice): a block for each class, teacher or room, and the master timetable, with the period times.
Save lesson list (CSV)	A plain list with one line for each lesson session, for sorting, filtering or other programs.

> **Tip:** the master timetable has many small boxes. It is easiest to read on A3.

8. Substitutions (cover for absent teachers)

Open **Day to day > Substitutions**. The timetable itself is never changed: absences and covers are kept separately, in the school file. Save the school when you are done.

1. **Add an absence**: choose the teacher, the first and last day (the same day for one day), and **The whole day** or **Only some periods** (tick them; they count on every day of the absence). A note is optional. To change an absence, remove it and add it again; removing it also removes its cover decisions.
2. Choose the **date** (or click an absence). The table lists each lesson the absent teachers would teach that day: period and times, class, subject, absent teacher and room.
3. For each lesson, press the orange button to **accept the suggestion**, **choose another teacher** from the list, or press **No cover needed** (for example, the class is out). **Clear** undoes the decision.

How the suggestion is chosen

- Only teachers who are **free** then: in that shift, not teaching, not on time off, not away themselves, not already covering another lesson, and not going over their **weekly maximum** (lessons plus covers that week).
- Teachers of the **same subject** come first, then the others. Within each group, the one with the **fewest covers** so far comes first.
- You can still choose a teacher who is not free. The page then shows a red warning that says why (for example "is teaching English (Grade 6 A) then").

Cover counts and the cover sheet

- **Cover counts** shows the total and the covers per teacher for the school year. **Reset counts...** (after a confirmation) starts counting again from today, for example at the start of a new school year. Earlier covers stay on the cover sheets.
- **Cover sheet**: choose the dates and press **Save PDF...** (one page per day, ready to print) or **Save CSV...** (for Excel).

9. If something goes wrong

What you see	What to do
"Please fix the problems on the Check page first"	Open **Check before building**. Each problem says how to fix it.
Some lesson sessions "could not be fitted in"	Read the reason under each one. Usually a teacher or class has too little free time. Free up time, or switch on *bending a few rules* in Rules & limits.
A red box when you drag	Hover it. It names the teacher, class or room that is in the way.
"Locked lessons are stopping other lessons from being placed"	The build page lists the locked lessons that are in the way. Press **Unlock these and build again**, or unlock some of them on the Timetable page.
Import says a row could not be imported	The message gives the file, the row and the reason. Fix that row and import again.
"The backup folder could not be found"	The USB stick or network drive is not connected, or the folder was moved. Connect it, or open **Backups** and choose the folder again. Your school is still saved in its own file.
"Not allowed to save files in the backup folder"	The folder is read-only for you. Choose another folder in **Backups**.
Names print as "?" in the PDF	The PDF supports Western letters only. Pictures (PNG) and Excel work with any letters.

Backups

Open **Backups** (under School setup) and press **Choose folder...**. Pick a USB stick, a shared drive, or a folder that OneDrive, Google Drive or Dropbox keeps in sync; Timewizer itself never goes online. Then:

- **Back up now** saves a copy of the school as it is on screen. Switch on **Back up every time I save** to get a copy with every save.
- Copies are named after the school file, with the date and time, for example *Riverside backup 2026-10-23 14-05-09.twz*. Timewizer keeps the newest 10 for each school (you can change the number) and removes older ones. Other files in the folder are never touched.
- **Restore from backup** lists the copies, newest first. **Open as new file...** asks for a new file name and opens the copy there. The school you have open is never replaced. When no school is open, the welcome screen has **Restore from backup...** too.
- The backup settings belong to the computer, not the school file.

Good habits

- Save often, and use **Save as...** to keep a copy before big changes.
- Choose a backup folder on another drive, and switch on the automatic backup.
- Lock the lessons you are happy with before building again.
- Build with a longer "time spent improving" (Rules & limits) for the final timetable.

Timewizer is open source (Apache-2.0).